Inference Host Proposal

Medical-LLM Glioma Classification Benchmark

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Request: Switching the medical-model runs from Featherless.ai to a dedicated host

Summary

The four open-weight medical models in our glioma benchmark (OpenBioLLM-70B, Med42-v2-70B, MedGemma-27B and Baichuan-M2-32B) are currently served through **Featherless.ai**, a flat-rate, shared serverless host. These lower-traffic models are repeatedly hitting **capacity limits** mid-run, which stalls and partially fails our 646-case batch. I propose adding **DeepInfra** (pay-as-you-go, dedicated GPU) so the benchmark can run end-to-end reliably, at an estimated **one-time cost of roughly $50–$100**.

What is happening

Featherless hosts models on a **shared serverless pool** under a flat monthly subscription. Because our medical models are not high-traffic, Featherless does not keep them loaded and warm — they are spun up on demand and share GPU capacity across all of Featherless's customers. When that pool is full, the API returns:

```
[182/646]  BraTS2021_01104
CAPACITY  503 model at capacity (wait 1/8) -- sleeping 20s
CAPACITY  503 model at capacity (wait 2/8) -- sleeping 40s
```

Our pipeline handles this gracefully — it waits and retries with increasing back-off (the `wait 1/8 … sleeping 20s/40s` lines above). But across 646 cases this causes long stalls, and some cases are skipped and must be re-run. **This is a host-side capacity limitation, not a defect in our code or data** — on a shared subscription we have no guaranteed throughput.

Proposal: add DeepInfra access

DeepInfra lets us run these same open-weight models on a **dedicated deployment with guaranteed capacity** — no shared-pool queueing, so the 503 stalls disappear and the full cohort runs in one pass. Two points for budgeting:

- **It is pay-as-you-go, not a subscription.** For these models we stand up a dedicated GPU deployment billed by runtime, and delete it the moment the run finishes — cost is bounded to the hours actually used.
- **It is the same code.** Our scripts already support DeepInfra as a provider; we simply point them at the deployment, change nothing in the prompt, schema, or scoring.

Estimated cost

DeepInfra's 2026 dedicated rates are **A100 $0.89/hr** and **H100 $1.79/hr**, billed only while the GPU is running. The table below assumes one metadata pass over all 646 cases, **sequential (single-stream) execution** (our scripts send one request at a time), 70B models on 2×H100 at full precision, smaller models on 1×H100, plus a 20% buffer for cold-starts and retries. These are deliberately conservative — an upper bound.

Model	Size	GPU (assumed)	$/hr	Est. GPU-hrs	Est. cost
OpenBioLLM-70B	70B	2 x H100	$3.58	9.7	~$35
Med42-v2-70B	70B	2 x H100	$3.58	9.7	~$35
MedGemma-27B	27B	1 x H100	$1.79	5.4	~$10
Baichuan-M2-32B	32B (reasoning)	1 x H100	$1.79	13.0	~$23
Total (one pass, all four)				**~38**	**~$100**

Figures are estimates based on conservative single-stream latency; actual cost depends on DeepInfra's serving configuration.

Levers that lower the cost

- **Use A100s instead of H100s** (~half price): total drops to roughly $50.
- **Run requests concurrently:** could cut wall-clock 3–5×, so real cost likely lands well under these figures.
- **FP8 single-GPU for the 70Bs** saves ~$35 — but for a publishable benchmark we should keep full precision.
- **Both metadata variants** (btreport + btreport_pp) roughly double the figure, to the $100–$200 range.

Recommendation and ask

Approve a **DeepInfra account and payment method** so we can stand up dedicated endpoints for the four medical models, complete the benchmark reliably, and tear them down afterward. The realistic commitment is a **one-off ~$50–$200 of usage**, not a recurring fee. I suggest a short **pilot of ~20 cases per model** first to confirm real-world latency and refine this estimate to the dollar before launching the full batch.

References

DeepInfra dedicated GPU pricing (A100/H100, 2026): deepinfra.com/pricing
DeepInfra Custom-LLM / private deployment billing: docs.deepinfra.com/private-models/custom-llms
Featherless serverless model hosting (shared pool, subscription): featherless.ai